

Prime Structure — Math-Clean Unified Specification (48/96/256)

Version: 1.0

Status: Normative

0. Scope

A single, mathematically rigorous specification that unifies the structure, gauge, and schedule layers. Targets three invariants:

- **48/256/12 288**: boundary subgroup sizes and minimal basis size.
- **R96**: exactly 96 distinct normalized responses.
- **3/8 compression**: canonical reduction from 256 selectors to 96 classes.

All claims below are self-contained and testable.

1. Core objects

1.1 Indexing and schedule

- Let $I = \mathbb{Z}/768$.
- Let $C = \{0, 1, 2\}$ be the channel set.
- Let $S = \mathbb{Z}/256$ be the selector set.

A **fair schedule** is a bijection $\sigma : I \rightarrow S \times C$. View $\sigma(i) = (s(i), c(i))$.

Fairness constraints. For every $s \in S$ and $c \in C$:

$$\#\{i \in I : s(i) = s\} = 3, \quad \#\{i \in I : c(i) = c\} = 256.$$

Equivalently, for any function $f : S \rightarrow \mathbb{R}$:

$$\frac{1}{768} \sum_{i \in I} f(s(i)) = \frac{1}{256} \sum_{s \in S} f(s),$$

and for any $g : C \rightarrow \mathbb{R}$:

$$\frac{1}{768} \sum_{i \in I} g(c(i)) = \frac{1}{3} \sum_{c \in C} g(c).$$

These equalities imply equal means. Equal variances follow by computing second moments under the same counting identities. Rotation of I preserves the constraints.

1.2 Boundary subgroup

Let

$$G^\circ = U(48) \times U(256), \quad |U(48)| = 16, |U(256)| = 128, |G^\circ| = 2048.$$

Concrete generators: - $U(48) \cong C_2 \times C_2 \times C_4$ generated by $\{-1, 5, 17\} \pmod{48}$.
- $U(256) \cong C_2 \times C_{64}$ generated by $\{-1, 5\} \pmod{256}$.

Action. G° acts on $S \times C$ by

$$(a, b) \cdot (s, c) = (a s \bmod 256, c), \quad a \in U(256), b \in U(48),$$

and on I through σ . Orientation is carried by the C factor and is fixed by the action.

2. R96 construction and 3/8 compression

2.1 Parameters

Let positive reals $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5, \alpha_6, \alpha_7, \alpha_8$ satisfy

$$\alpha_0 = 1, \quad \alpha_4 \alpha_5 = 1.$$

Define the bit index set $T = \{1, 2, 3, 6, 7, 8\}$. Define a tri-state **pair variable** $d \in \{-1, 0, 1\}$ capturing the relative choice of the unity pair $(4, 5)$.

2.2 Map and normal form

For $b \in \{0, 1\}^T$ and $d \in \{-1, 0, 1\}$ define

$$R(b, d) = \left(\frac{\alpha_4}{\alpha_5} \right)^d \prod_{i \in T} \alpha_i^{b_i}.$$

This is the pair-normalized form; α_0 drops out and α_4, α_5 enter only through their ratio.

2.3 Genericity hypothesis

Assume the multiset

$$\mathcal{L} = \{\log(\alpha_4/\alpha_5)\} \cup \{\log \alpha_i : i \in T\}$$

is linearly independent over $\mathbb{Q}$.

2.4 Main theorem (R96)

Under the hypothesis above, the map

$$\phi : \mathbb{Z}_3 \times \mathbb{Z}_2^5 \longrightarrow \mathbb{R}_{>0}, \quad \phi(d, \bar{b}) = R(b, d)$$

(where $\bar{b}$ denotes any representative of the 5-dimensional parity class of b) is injective. Hence

$$|\text{Im } R| = 3 \cdot 2^5 = 96.$$

Proof sketch. Take logs to embed into $\mathbb{R}$. The image is the subset of the lattice spanned by $\mathcal{L}$ with coordinates in $\{-1, 0, 1\} \times \{0, 1\}^5$. By $\mathbb{Q}$ -independence, distinct exponent vectors map to distinct reals.

2.5 Corollary (3/8 compression)

The canonical reduction $\kappa : S \rightarrow C_{96}$ obtained by composing the 256 selectors with $(d, \bar{b}) \mapsto R(b, d)$ yields exactly 96 equivalence classes. The compression ratio is $96/256 = 3/8$.

2.6 Concrete model of C_{96}

Define the commutative semiring

$$C_{96} \cong \mathbb{Z}_3 \times B^5,$$

where $B = \{0, 1\}$ with addition XOR and multiplication AND. Operations are componentwise. The embedding

$$\iota : \mathbb{Z}_3 \times \mathbb{Z}_2^5 \rightarrow C_{96}, \quad (d, \bar{b}) \mapsto (d \bmod 3, b)$$

followed by any monotone evaluation of α s realizes R as a semiring homomorphism after fixing an interpretation of products as conjunctions of active toggles. The **crush** map $C_{96} \rightarrow B$ given by

$$(d, \mathbf{b}) \longmapsto (d / 0) \ominus \mathbf{R} (\mathbf{b} / \mathbf{0}) =$$

is a semiring homomorphism onto the Boolean semiring.

3. Schedule layer (C768)

3.1 Closed forms

Let $\mathbf{1}$ be the all-ones vector on I . For any $h : S \rightarrow \mathbb{R}$ and $k : C \rightarrow \mathbb{R}$, define lifted functions on I by $h \circ s$ and $k \circ c$. Then

$$\langle h \circ s, \mathbf{1} \rangle = 3 \sum_{u \in S} h(u), \quad \langle k \circ c, \mathbf{1} \rangle = 256 \sum_{v \in C} k(v).$$

For second moments, with $h_1, h_2 : S \rightarrow \mathbb{R}$ and $k_1, k_2 : C \rightarrow \mathbb{R}$,

$$\begin{aligned} \sum_{i \in I} (h_1 \circ s)(i)(h_2 \circ s)(i) &= 3 \sum_{u \in S} h_1(u)h_2(u), \\ \sum_{i \in I} (k_1 \circ c)(i)(k_2 \circ c)(i) &= 256 \sum_{v \in C} k_1(v)k_2(v). \end{aligned}$$

Hence equal means and equal variances hold for any statistics aggregated along S or C . These identities are invariant under rotations of I .

3.2 Deterministic constructors

Provide any bijection σ generated by a pair of coprime strides (p, q) on S and C :

$$\sigma(t) = (pt \bmod 256, qt \bmod 3), \quad t \in I,$$

with $p \in U(256)$, $q \in U(3) = \{1, 2\}$. This meets all fairness identities.

4. Basis size 12 288

Let a **basis** be any minimal index set $\text{Idx} \subseteq I \times S \times C$ that generates all schedules under the G° action while preserving channel orientation and fairness.

Lower bounds. 1. Fairness implies $|\text{Idx}|$ is a multiple of $|C| \cdot |S| = 3 \cdot 256$.

2. The oriented boundary action forces $|\text{Idx}|$ to be a multiple of $|U(48)| = 48$ at the orbit level.

Taking $\text{lcm}(3 \cdot 256, 48) = 12\,288$ yields the universal lower bound.

Attainment. Construct Idx by fixing one representative per G° -orbit of $S \times C$ and one channel orientation. This meets the bound, hence $|\text{Idx}| = 12\,288$ is minimal.

5. Φ -module and torsor of lifts

5.1 Signatures

Let $\Phi : \{0, 1\}^{12} \rightarrow \{0, 1\}^{10}$ be linear over $\mathbb{F}_2$ with matrix $S \in \mathbb{F}_2^{12 \times 10}$. Use the explicit full-rank matrix (rank 10):

$$S = \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Then $\ker S$ has size $2^{12-10} = 4$. For any signature $y \in \{0, 1\}^{10}$, the fiber $\Phi^{-1}(y)$ is a **torsor** under $\ker S$.

5.2 NF-Lift

Define a canonical lift $\text{lift} : \{0, 1\}^{10} \rightarrow \{0, 1\}^{12}$ by solving $Sx = y$ with Gaussian elimination over $\mathbb{F}_2$ and selecting the lexicographically smallest representative in the affine coset. This is well-defined because each fiber is nonempty and finite.

Round-trip property. $\Phi(\text{lift}(y)) = y$ for all y . **Stability.** Composition with any $k \in \ker S$ leaves signatures invariant: $\Phi(x \oplus k) = \Phi(x)$.

6. Conformance

All MUST-level tests are reproducible with the data below.

6.1 Boundary tests

- Verify generators: orders $\text{ord}_{48}(-1) = 2$, $\text{ord}_{48}(5) = 4$, $17 \equiv 1 \pmod{16}$, $17 \equiv 2 \pmod{3}$.
- Verify $\text{ord}_{256}(-1) = 2$, $\text{ord}_{256}(5) = 64$.
- Enumerate $|U(48)| = 16$, $|U(256)| = 128$, product 2048.

6.2 Schedule tests

- Check bijection of σ .
- Count identities for means and variances using §3.1 formulas.
- Rotation invariance: for any $r \in \mathbb{Z}/768$, $i \mapsto i + r$ leaves the statistics unchanged.

6.3 R96 tests

- Evaluate $R(b, d)$ for all $d \in \{-1, 0, 1\}$, $b \in \{0, 1\}^T$; assert cardinality 96.
- Confirm pair normalization: replacing (α_4, α_5) by $(t\alpha_4, t^{-1}\alpha_5)$ leaves all values unchanged.

6.4 Basis tests

- Using the group action, generate schedules from Idx and count $|\text{Idx}| = 12\,288$.
- Shrink attempts below 12 288 must fail by violating at least one lower-bound condition of §4.

6.5 Φ -module tests

- Compute rank of S and $|\ker S| = 4$.
- Verify torsor action on each fiber.
- Verify NF-Lift round-trip and lexicographic minimality.

7. Deterministic artifacts

- **Schedule constructor:** $\sigma(t) = (5t \bmod 256, 2t \bmod 3)$.
- **Boundary generators:** as in §1.2.
- **NF-Lift solver:** fixed pivot order $(1, 2, \dots, 10)$, ties broken by smallest index first.
- **Replay seed:** integer seed applied to any pseudorandom choices outside this spec must be recorded and produce identical acceptance outcomes.

8. Notes on implementation

- Use modular arithmetic libraries that expose unit groups and orders.
- For Φ , implement Gaussian elimination over $\mathbb{F}_2$ with bit-packed rows.
- For R96, treat `\nabla\log` parameters as real vectors; independence should be enforced at configuration time via rank checks over $\mathbb{Q}$ -tests on rational relations or via randomized Diophantine screening.

This document is complete for verification of the 48/96/256 invariants, the R96 map, the 3/8 compression, the schedule fairness identities, the basis size 12 288, and the Φ -module torsor with NF-Lift.